

WORKSHOP

Not Every Agent Needs a Frontier Model

CYPHER 2026 | Nitin Agarwal

What we will cover

01

Stage classification

SLM-appropriate stages (classification, extraction, routing, reranking) vs. LLM stages (synthesis, ambiguity, judgment).

02

Hybrid pipeline design

Explicit per-stage model selection — not “call the biggest model for everything.”

03

Passive RAG → agentic

Query rewriting, multi-agent routing, optional self-reflection — and whether the extra reasoning is worth it.

04

Real metrics

Quality, latency (P50/P95), cost per query, SLM-vs-LLM call mix — not just a plausible-looking answer.

The default is expensive

Most multi-agent systems route every call to the largest model — maximizing quality on hard tasks and wasting budget on easy ones.

Latency

Latency

Frontier models add seconds per call. Classification and extraction rarely need that depth.

Cost

Cost

Token price scales with model size. Easy stages often dominate total call volume.

Capacity

Capacity

GPU and rate limits are finite. Spending them on rewrite steps starves judgment stages.

Thesis: match model size to stage difficulty — then measure it.

Which stages need a frontier model?

SLM-APPROPRIATE

Small models are enough

Classification

Intent, validity, topic tags

Extraction

Keywords, entities, fields

Routing

Phase / agent handoff

Reranking

Lightweight relevance

Constrained rewrite

Short keyword queries

LLM-REQUIRED

Still need stronger models

Open-ended synthesis

Plans, long reports

Ambiguity resolution

Conflicting evidence

Multi-criteria judgment

AGI scoring (10 axes)

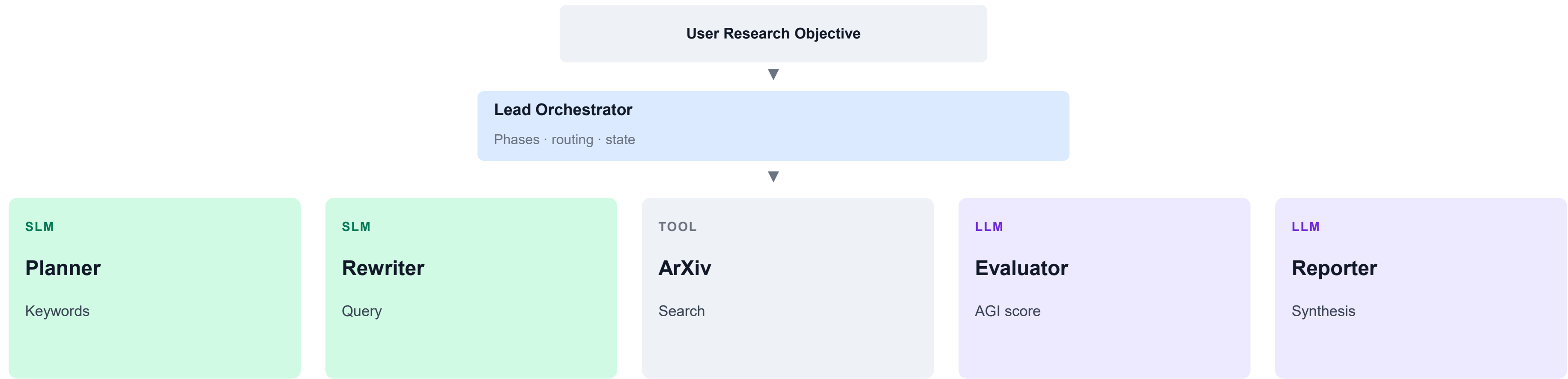
Self-reflection

Critique / gap analysis

Nuanced ranking

Subtle quality differences

How the agents connect



Three modes on the same graph

SLM-only	LLM-only	Hybrid
All stages → 0.5B Fastest, cheapest	All stages → 8B Highest quality	Stage-aware routing Recommended default

Measured results (FIXED run)

SLM wall-clock

52s

6 calls · 0 LLM

LLM wall-clock

266s

6 calls · 0 SLM

Hybrid wall-clock

121s

2 SLM · 4 LLM

LLM avg AGI

71.1

2 High papers

Mode	Papers	Avg AGI	High	Wall-clock	SLM	LLM	Est. cost
SLM	3	61.1	1	52s	6	0	\$0.00016
LLM	3	71.1	2	266s	0	6	\$0.00132
Hybrid	3	55.1	0	121s	2	4	\$0.00087

Key read

SLM is ~5× faster than pure LLM. Hybrid cuts latency roughly in half vs LLM while keeping judgment on the large model (4 of 6 calls). Quality edge still sits with the LLM — as expected for multi-criteria scoring — but cost and latency favor Hybrid for production volume.

Design principles

1

Classify stages first

Label each step as classification / extraction / routing vs. judgment / synthesis before picking a model.

2

Default to SLM, escalate to LLM

Use the small model unless the stage requires open-ended reasoning or multi-criteria judgment.

3

Make routing explicit

A `choose_model(stage, mode)` policy beats “always call the biggest model.”

4

Measure the mix

Track quality, P50/P95 latency, cost, and SLM vs LLM call share — not just whether the answer looks good.

5

Agentic only when it pays

Query rewrite and multi-agent routing help when discovery is fragile; pure judgment stages still need the LLM.

Not every agent needs a frontier model.

Match model size to stage difficulty.
Measure quality, latency, cost, and call mix.